Exploring Interaction Between Behavior Cloning and Reward Shaping in Chess

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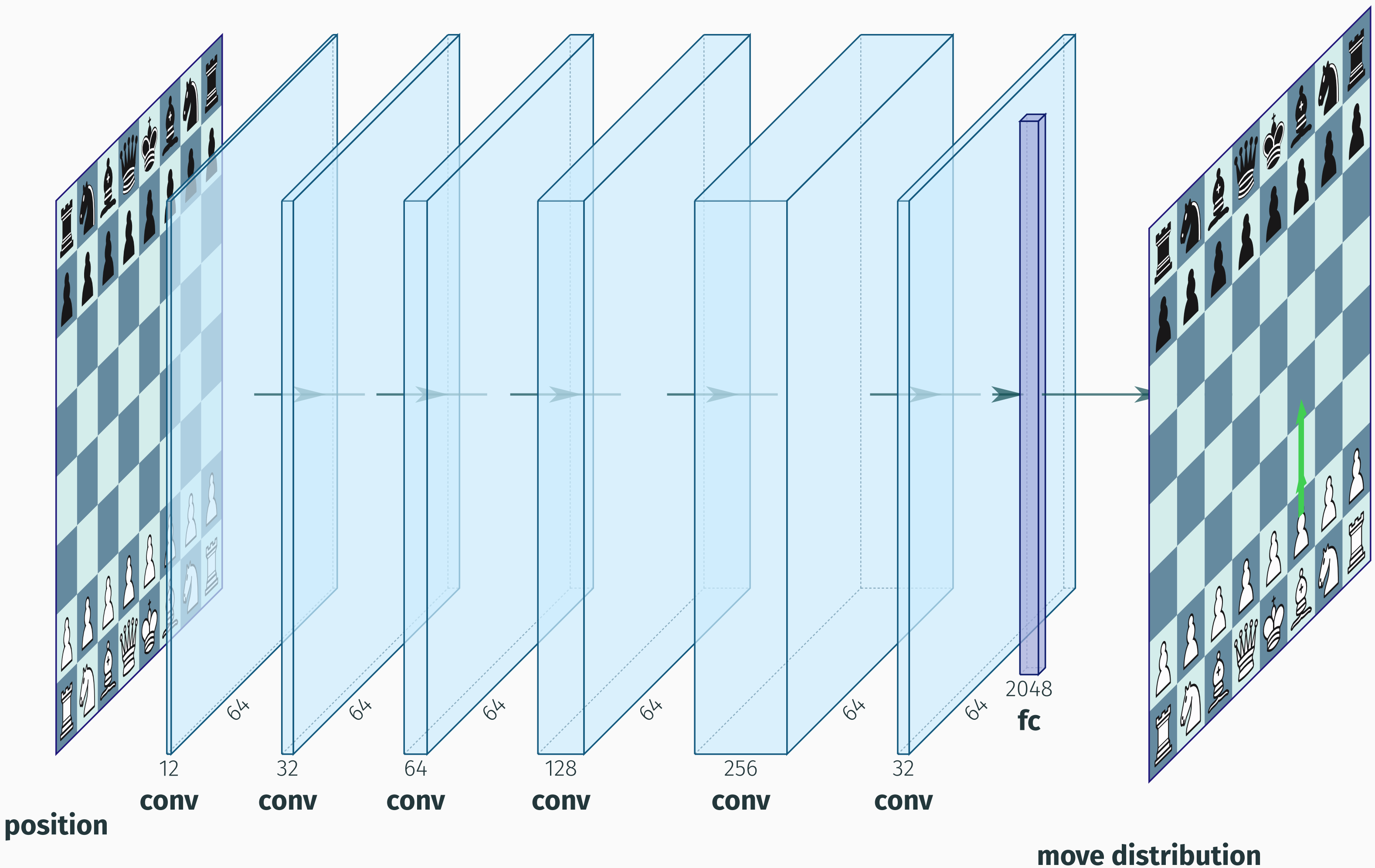
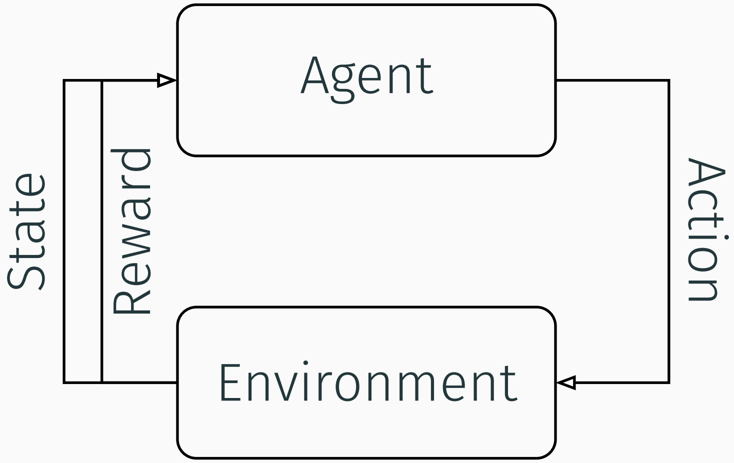
Introduction



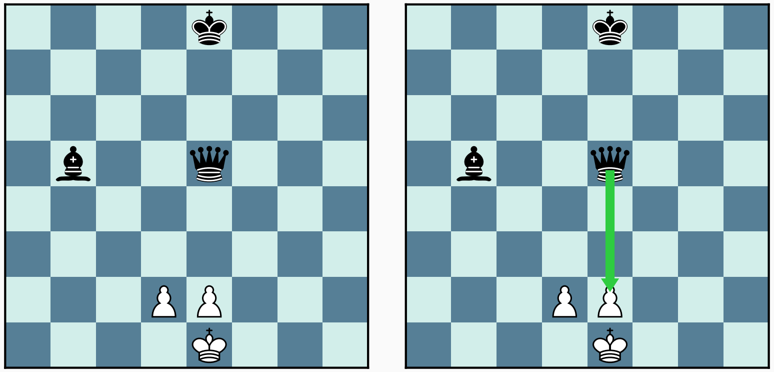
Reinforcement Learning



Problems

- Sparse rewards → poor performance
- Reward Shaping → difficult:
  - Missaligned rewards
  - Reward hacking
  - Prioritizing wrong rewards

Imitation Learning: Behavior Cloning



- Supervised learning
- Agent learns expert demonstrations
- e.g. puzzles, grandmaster games

Problems

- Demonstration sample bias → poor performance

Combine both approaches

1. Pretrain: Behavior Cloning
2. Finetune: Reward Based Reinforcement Learning

Can both approaches stabelize each other?

- Behavior Cloning: Stabelize RL with sparse rewards
- Reinforcement Learning: Compensates for data bias

Lets find out!

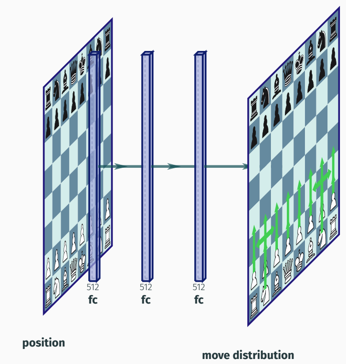
- Train different models:
  - Architecture: Fully connected, cnn, residual block
  - Pretraining Data: Heavily biased → less biased
  - Rewards Sets: Sparse → Dense

Architecture

Architecture

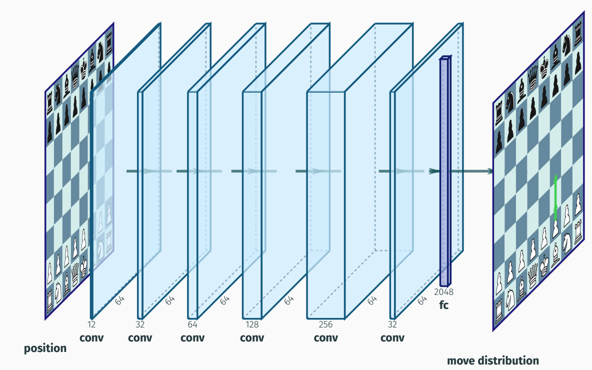
FCNN

- 3 hidden layers: 512 neurons



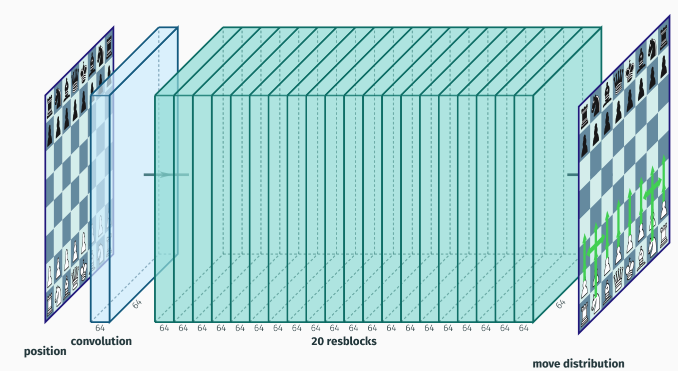
CNN

- increasing number of channels:
- $12 \to 32 \to 64 \to 128 \to 256 \to 32$



ResNet

- 20 blocks with 64 channels

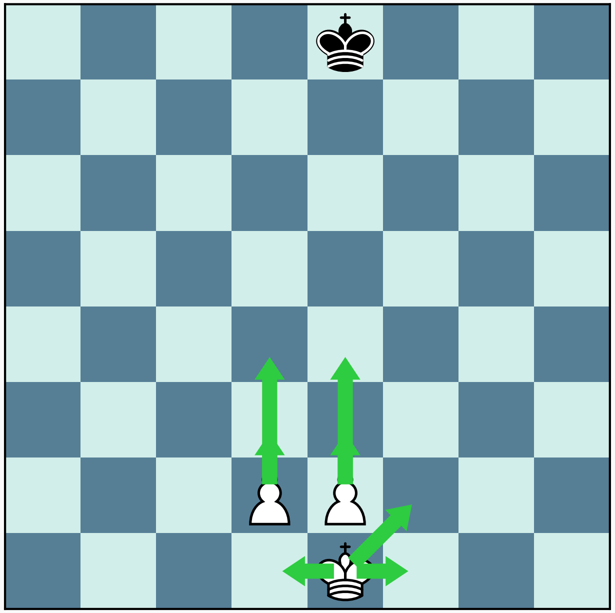


Input

- tensor `shape = (8,8,12)`:
  - 8×8 squares
  - 12 pieces: $\{\text{White}, \text{Black}\} \times \{\text{Pawn}, \text{Rook}, \text{Knight}, \text{Bishop}, \text{Queen}, \text{King}\}$

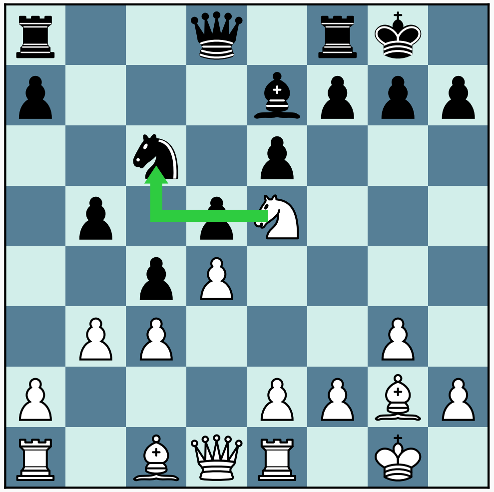
Output

- Move distribution: from square → to square
- **But:** Not all moves are legal → mask illegal moves



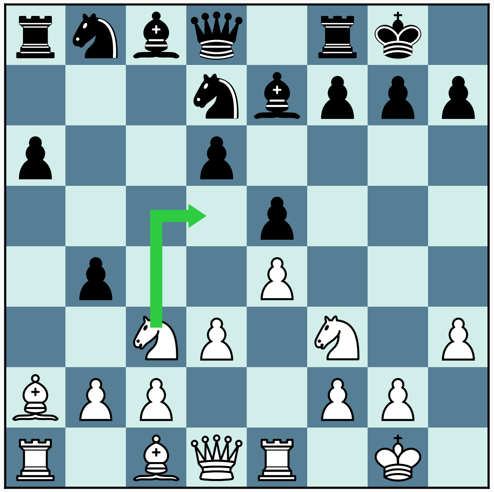
Imitation Learning

Strategy vs Tactics



Tactics

- Short term moves
  - winning a piece
  - checkmate
  - etc.



Strategy

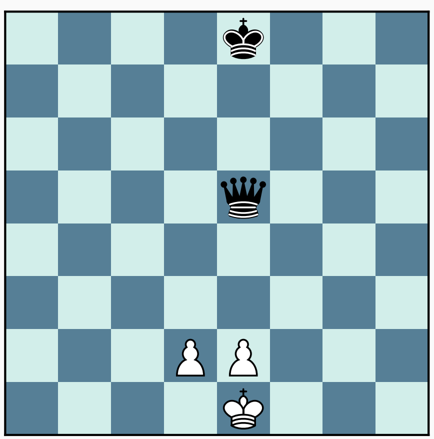
- Long term move:
  - Good piece positioning
  - Pressure on oponent King
  - Control of center
  - etc.

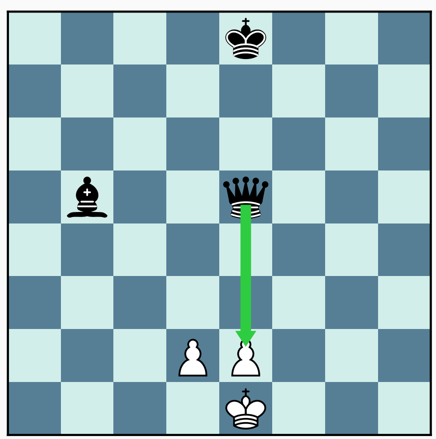
Lichess Puzzles

- 6 million positions
- tactical positions

Bias

- no strategic positions
- no positions without tactic



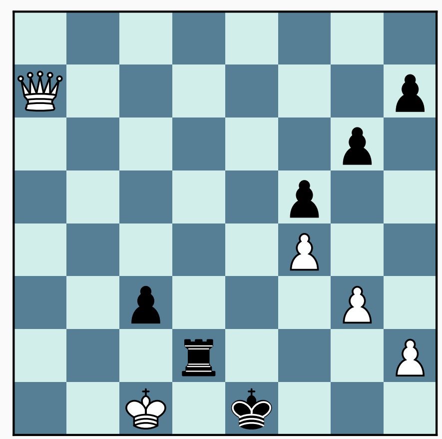


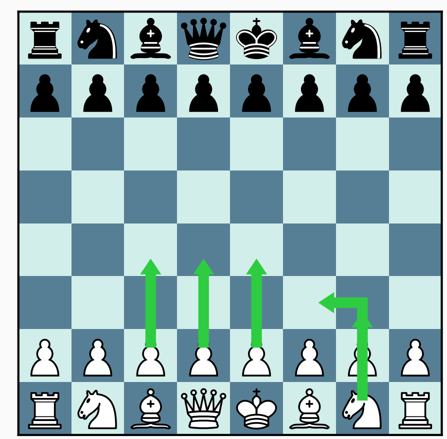
GM games

- chess.com master games
- balance tactics/strategy

Bias & Problems

- high level play
- many resignations
- multiple moves possible





Reinforcement Learning

Reward sets

0. **Sparse:** rewards only for true goal
   - reward for win/loss/draw
1. **Medium:** more rewards, but might misalign with goals
   - control center, material win, king safety
2. **Dense:** highest risk of misalignement
   - control outer center, control center, castling, promoting, blunder prevention, material, king safety, give check, win

Total Models

Training

- Architecture: $\{\mathrm{FCNN}, \mathrm{CNN}, \mathrm{ResNet}\}$
- Pretraining: $\{\mathrm{Puzzles}, \mathrm{Masters}\} \times \{1\ \mathrm{Epoch}, 10\ \mathrm{Epochs}\}, \{\mathrm{Untrained}\}$
- Reward Set: $\{\mathrm{R\_0}, \mathrm{R\_1}, \mathrm{R\_2}\}$
  - 1000 batches with 16 games each (REINFORCE)

→ 57 models

Results

Behavior Cloning: Training Accuracy

dataset	epochs	fc	cnn	resnet
puzzle	1	62.0%	69.9%	74.6%
	10	65.8%	73.3%	81.7%
grand masters	1	31.0%	38.5%	41.5%
	10	31.8%	39.6%	43.7%

- Puzzle better generalizing
- **But** master games: different labels, same position

Evaluation

- All models plays 1000 games against all other model (same architecture)
- Normalized tournament score (win 1 point, draw 0.5 points, loss 0 points)

BC model with RL

dataset	epochs	fc	cnn	resnet
none		0.310	0.251	0.405
puzzle	1	0.615	0.469	0.515
	10	0.562	0.538	0.533
grand masters	1	0.472	0.556	0.510
	10	0.494	0.624	0.513

- Models improve significantly with pretraining

Reward Shaping

RL models with Pretraining

rewards	fc	cnn	resnet
none	0.461	0.419	0.855
r_0	0.520	0.547	0.404
r_1	0.504	0.532	0.405
r_2	0.507	0.486	0.407

RL models without Pretraining

rewards	fc	cnn	resnet
r_0	0.289	0.266	0.408
r_1	0.307	0.235	0.408
r_2	0.335	0.251	0.399

- Depending on the model RL hurts performance
- Sparse rewards generally better

Conclusion

- We could not find any training strategy across architectures
- Overall denser rewards hurt model performance

Possible Improvements

- **Data Prep:** Unify data in Masters games
- **Architecture:** Hyperparameter tuning on pretrain accuracy
- **BC:** Mix of master games and Puzzles
- **RL:** Longer training
- **RL:** Iterative reward shaping

How does it really play?

Well.... https://lichess.org/cuRJkeJ1#0